Predicting Body Fat Percentage

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This study investigates whether body fat percentage, a key indicator of health, can be accurately predicted using simple and accessible body measurements. Using a dataset of physical characteristics, a multiple linear regression model was developed to identify the most relevant predictors. Feature selection methods were then applied, and candidate models were evaluated based on out-of-sample performance. The final model, consisting of age, abdomen circumference, wrist circumference, and height, achieved the best balance between predictive accuracy and simplicity. It explains approximately 73% of the variation in body fat percentage. Overall, the findings suggest that body fat percentage can be reliably estimated using a small set of easily obtainable measurements, offering a simple and practical predictive tool.

1. Introduction

Body fat percentage is a key indicator of health that provides more meaningful information than body weight alone. It is commonly used to assess fitness levels and health risks, but accurate measurement typically requires clinical methods such as DEXA scans, which are often costly and not easily accessible.

This study aims to determine whether body fat percentage can be accurately predicted using simple and easily measurable body characteristics. Specifically, it addresses the following questions: how body fat percentage is related to basic body measurement variables; which variables are the most important predictors of body fat percentage; and whether a parsimonious model can achieve strong predictive performance compared to more complex alternatives. These questions are important in assessing whether simple body measurements can serve as a practical and reliable alternative to more complex and costly measurement methods.

To investigate this, we use a dataset of physical characteristics as a training set and evaluate multiple linear regression models to determine which specification provides the best balance of predictive performance and simplicity.

2. Data Set

The bodyfat dataset contains 250 observations of men of various ages. The raw dataset consists of 16 variables, with body fat percentage ($pct.bf$) as the dependent variable. Aside from density, pct.bf, age, weight, and height, the remaining variables are body circumferences (in centimetres) measuring the perimeter around different body parts. A full summary of the variables is provided in Figure 2 in the appendix.

2.1. Exploratory Data Analysis. From the correlation matrix plot (Figure 2), we observe a near-perfect negative correlation (-0.99) between density and body fat percentage. This means we can predict body fat percentage solely from density. However, we note that density is calculated by dividing body mass (weight) by body volume. Although weight is easy to obtain, measuring body volume requires specialised methods such as water displacement, which are costly and not easily accessible. Hence, we decided to drop density from our dataset. (Figure 2) also shows a perfect positive correlation between waist and abdomen, indicating that these variables are effectively interchangeable. So we removed waist and retained abdomen.

3. Analysis

3.1. Data Transformation. Height was recorded in inches while all other body measurements (chest, abdomen, and so on) were in centimeters, so height was converted to centimeters for consistency. Weight was converted from pounds to kilograms to align with standard metric units commonly used. We observe that the dependent variable, body fat percentage, has one observation with value 0, which we omitted and identified as measurement error since this is not scientifically possible (Ebeling, 2020).

3.2. Initial Model Experimentation. We start by fitting a linear model using all predictors except waist. The residuals vs fitted plot (Figure 3) shows a mild wave-like pattern showing slight overestimation at the lower and upper ends. However, the deviations are small and likely driven by sparse data (at upper end) rather than genuine nonlinearity. The linearity assumption is considered reasonably satisfied.

The model achieves an R^2 value of 0.7466 ($Adjusted\ R^2 = 0.7326$), though only age, abdomen, and wrist are statistically significant. Given the strong correlations between predictor variables observed in EDA, we check for multicollinearity using VIF. Weight stands out with a VIF over 40, while being heavily correlated with other predictors (see Figure 2). Furthermore, weight does not distinguish between fat and lean mass, so its inclusion may inflate model fit without adding meaningful predictive value. Weight is thus excluded.

Refitting without weight keeps R^2 unchanged but raises $Adjusted\ R^2$ to 0.734. Chest, abdomen, hip, and thigh still show VIF > 5, as they measure overlapping body parts. We retain abdomen over chest and hip over thigh based on stronger correlations with body fat. So chest and thigh are dropped, confirmed by their high p-values. Upon refitting, hip continues to show high VIF with abdomen and remains insignificant, so it is dropped too.

Knee and ankle are removed at this stage. Intuitively, these are joints whose size reflects bone structure rather than fat accumulation, confirmed by their high p-values. Bicep and forearm are unreliable fat predictors as someone with large biceps could have either low (muscular) or high (larger person) body fat, confirmed by their high p-values. Neck, while borderline significant ($p = 0.058$), becomes insignificant once bicep and forearm are removed, suggesting its earlier significance was partly driven by surrounding noise. All three are dropped, giving our first candidate model (age + abdomen + wrist + height_cm). An ANOVA test comparing the full model against the reduced model confirms that the dropped predictors do not meaningfully improve fit ($F = 1.43$, $p = 0.185$).

3.3. Model Comparison. After manually arriving at a candidate model, we also identified candidate models through three feature selection approaches:

3.3.1. Exhaustive Search. We first perform an exhaustive search, which considers all possible combinations of predictors. Each model is evaluated using BIC (Bayesian Information Criterion), which, compared to the Akaike Information Criterion, imposes a heavier penalty on model complexity and therefore tends to favour simpler, more parsimonious models. As this aligns with our objective, we adopt BIC as our selection criterion.

Based on Figure 4, we observe that BIC reaches its minimum when the model includes 4 predictors. Therefore, we select the model with age, height, abdomen, and wrist (Figure 5). Interestingly, we obtain the same model as the first candidate model.

3.3.2. Backward Variable Selection. We also conducted backward variable selection based on BIC. In this approach, variables are removed one at a time until the model reaches its minimum BIC value. Based on Figure 6, the model achieves the lowest BIC when age, height, abdomen, and wrist are included as predictors. Notably, this also results in the same set of predictors as those obtained from the manual and exhaustive search methods.

3.3.3. Forward Variable Selection. For further comparison, we also performed forward variable selection based on BIC. In this approach, variables are added one at a time until the model reaches its minimum BIC value. According to Figure 7, the resulting model includes abdomen, weight, and wrist as predictors, which becomes our second candidate model.

To assess which candidate model is best, we perform out-of-sample tests next.

3.4. Cross Validation. To evaluate the out-of-sample performance on unseen data, we apply 10-fold cross-validation across our 2 candidate models. This repeatedly partitions the data into 10 equal folds, training on 9 and testing on the remaining 1, cycling through all folds.

We tested this approach on 2 different models:

- Model 1: Manual/Exhaustive search/Backward selection

$$\widehat{pct.bf} = 3.69 + 0.056\,age + 0.767\,abdomen - 1.929\,wrist - 0.128\,height_cm \quad [1]$$

- Model 2: Forward variable selection

$$\widehat{pct.bf} = -27.259 + 0.957\,abdomen - 0.208\,weight - 1.388\,wrist \quad [2]$$

Both models perform similarly in terms of predictive ability. Based on MAE and RMSE (see Figure 8), Model 1 performs slightly better than Model 2, with MAE lower by 0.034 and RMSE lower by 0.031. This suggests Model 1 produces more accurate predictions on average and is slightly better at limiting large prediction errors.

In contrast, Model 2 has a marginally higher R^2 (0.745 vs 0.724), indicating that it explains a slightly greater proportion of the variation in body fat percentage. However, a higher R^2 does not necessarily imply better predictive performance, especially in this case. The improvement in R^2 is likely driven by the inclusion of weight, which we identified as a problematic variable due to multicollinearity and weak relevance.

Importantly, both exhaustive and backward selection procedures (based on BIC) converge to the same specification as Model 1. This provides additional evidence that a parsimonious specification can achieve comparable, if not better, predictive performance. Therefore, Model 1 is preferred as the more parsimonious and robust model, retaining only the most reliable predictors while avoiding unnecessary complexity.

3.5. Assumptions Checking. In Figure 9, the residuals vs fitted values plot for the final model shows a mild wave-like pattern, similar to Figure 3. However, as discussed previously, this is likely driven by limited observations at the upper extremes rather than true non-linearity, so the linearity assumption is considered satisfied. The Q-Q plot is approximately linear with no major violations of normality, and the scale-location plot shows a roughly horizontal trend with no clear evidence of heteroskedasticity (Figure 10). Independence is assumed as each observation corresponds to a different individual with no repeated measurements.

Finally, no observations exceed three standard deviations from the mean, suggesting that there are no significant outliers.

4. Discussion and Conclusion

The final model achieved an R^2 value of 0.73, meaning that 73% of the variation in body fat percentage is explained by the model.

Abdomen is the strongest predictor, with a coefficient of 0.77. This means a 1 cm increase in abdomen circumference is associated with a 0.77 increase in body fat percentage, holding other variables constant. Age showed a positive association, since people become less active and their bodies change composition as they get older. Height has a negative association, likely because taller individuals distribute mass over a larger frame, resulting in lower body fat.

Wrist circumference had a negative coefficient (-1.93), suggesting that larger wrists are associated with lower body fat. Interestingly, we observed that wrist showed a slight positive correlation with body fat in isolation, but reverses direction in the full model - an example of Simpson's paradox. Future work should investigate this further by exploring alternative specifications and additional predictors to better understand the true role of wrist circumference in predicting body fat percentage.

The 95% prediction interval plot (Figure 11) shows predictions closely follow the diagonal, indicating good overall fit. However, the interval is relatively wide and dips into negative values at the lower end, which is not physically possible. This suggests reduced reliability for individuals with very low body fat. Future work could explore bounded models such as beta regression.

Key limitations include: First, the sample dataset includes only adult males, so model results may not generalise well to other populations. Second, while the sample size is adequate, it may be insufficient to reliably capture patterns at the extremes of the distribution. Finally, the predictor variables rely on manual tape measurements, potentially introducing slight measurement error.

Future research could address these by including more diverse populations, larger datasets, and additional predictors such as physical activity and diet. Overall, the model shows that body fat percentage can be reasonably estimated from a small set of body measurements, with relatively good predictive power.

$$\widehat{pct.bf} = 3.69 + 0.056\,age + 0.767\,abdomen - 1.929\,wrist - 0.128\,height_cm \quad [3]$$

References

Ebeling A (2020). "Ronnie Coleman's Body Fat." Accessed: 19 April 2026, URL https://www.menshealth.com/health/a33247811/ronnie-coleman-body-fat/.

A. Appendix

Variable	Description	Unit
Density	Body density	g/cm^3
Pct.BF	Body fat percentage	%
Age	Age of individual	years
Weight	Body weight	lb
Height	Height	in
Neck	Neck circumference	cm
Chest	Chest circumference	cm
Abdomen	Abdomen circumference	cm
Waist	Waist circumference	in
Hip	Hip circumference	cm
Thigh	Thigh circumference	cm
Knee	Knee circumference	cm
Ankle	Ankle circumference	cm
Bicep	Bicep circumference	cm
Forearm	Forearm circumference	cm
Wrist	Wrist circumference	cm

Fig. 1. Table Summary of Dataset

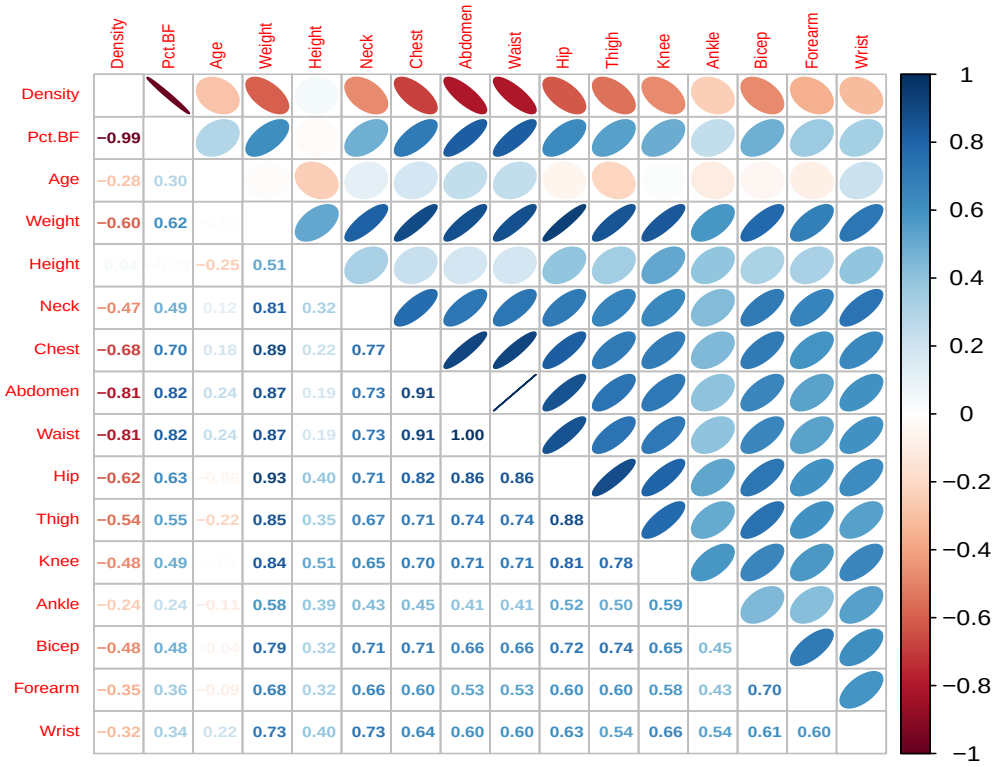

Fig. 2. Correlation Matrix Plot

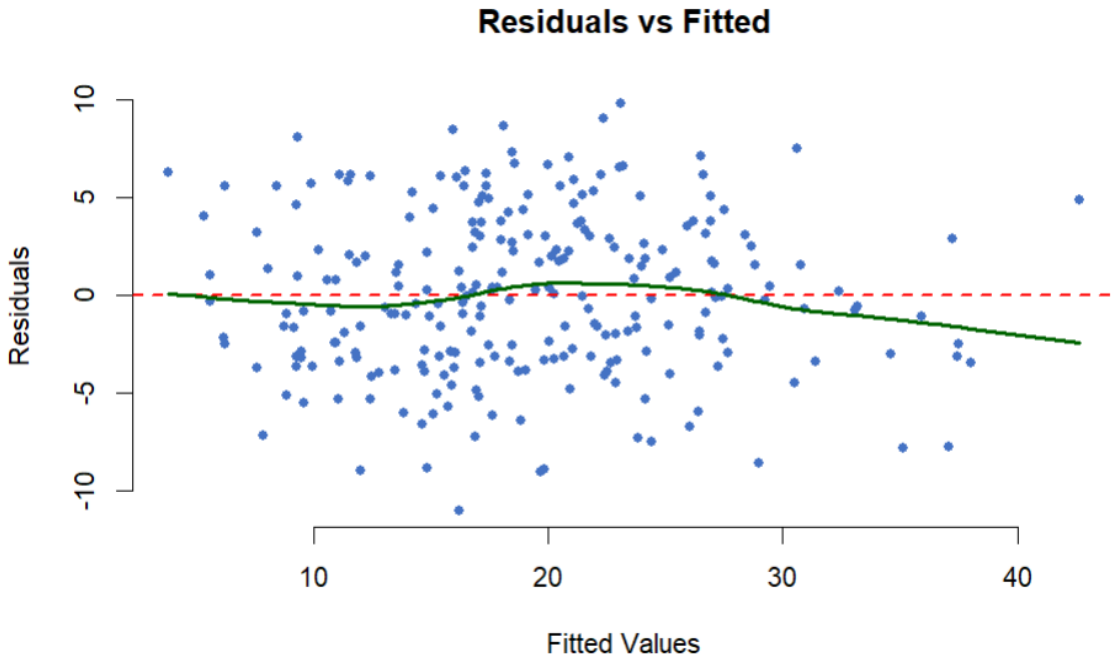

Fig. 3. Residual vs Fitted Plot

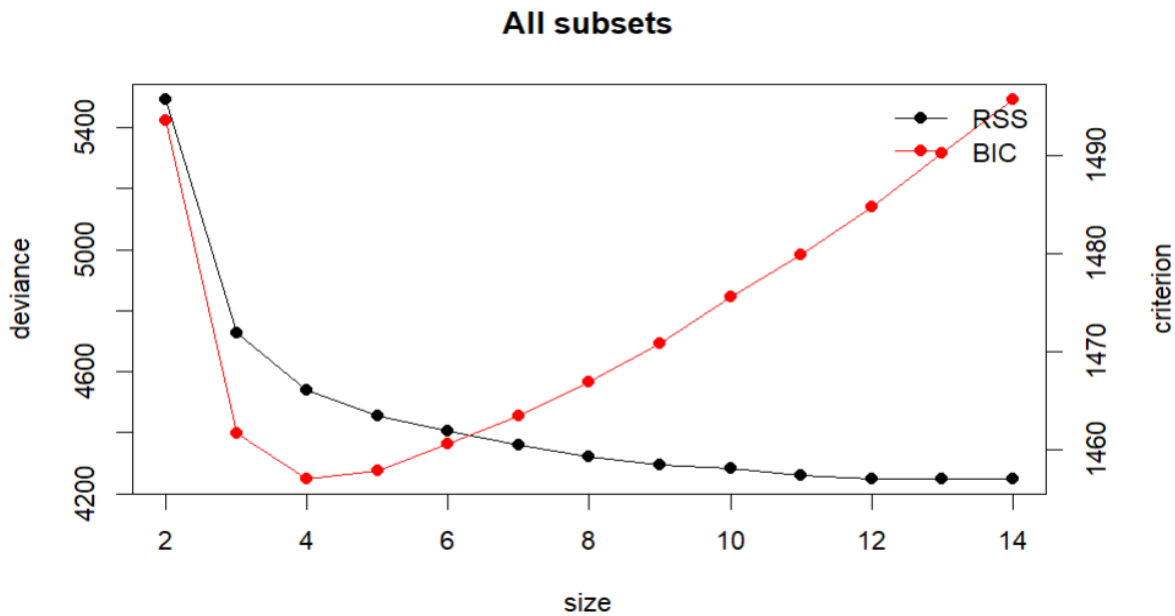

Fig. 4. Exhaustive Search

		age	neck	chest	abdomen	hip	thigh	knee	ankle	bicep	forearm	wrist	weight_kg	height_cm
1	(1)	" "	" "	" "	"*"	" "	" "	" "	" "	" "	" "	" "	" "	" "
2	(1)	" "	" "	" "	"*"	" "	" "	" "	" "	" "	" "	" "	"*"	" "
3	(1)	" "	" "	" "	"*"	" "	" "	" "	" "	" "	" "	"*"	"*"	" "
4	(1)	"*"	" "	" "	"*"	" "	" "	" "	" "	" "	" "	"*"	" "	"*"
5	(1)	"*"	" "	"*"	"*"	" "	" "	" "	" "	" "	" "	"*"	" "	"*"
6	(1)	"*"	" "	"*"	"*"	" "	" "	" "	" "	"*"	" "	"*"	" "	"*"
7	(1)	"*"	"*"	"*"	"*"	" "	" "	" "	" "	"*"	" "	"*"	" "	"*"
8	(1)	"*"	"*"	"*"	"*"	" "	" "	" "	" "	"*"	"*"	"*"	" "	"*"
9	(1)	"*"	"*"	"*"	"*"	"*"	" "	" "	" "	"*"	"*"	"*"	" "	"*"
10	(1)	"*"	"*"	"*"	"*"	"*"	"*"	" "	" "	"*"	"*"	"*"	" "	"*"
11	(1)	"*"	"*"	"*"	"*"	"*"	"*"	" "	"*"	"*"	"*"	"*"	" "	"*"
12	(1)	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	" "	"*"
13	(1)	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"	"*"

Fig. 5. Exhaustive Search (Variables Selection)

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.690	8.102	0.456	0.649
age	0.056	0.024	2.365	0.019
abdomen	0.767	0.034	22.751	0.000
wrist	-1.929	0.409	-4.712	0.000
height_cm	-0.128	0.048	-2.673	0.008

Fig. 6. Backward Variable Selection

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-27.259	6.743	-4.043	0.000
abdomen	0.957	0.055	17.334	0.000
weight_kg	-0.208	0.053	-3.897	0.000
wrist	-1.388	0.435	-3.186	0.002

Fig. 7. Forward Variable Selection

Model	RMSE	MAE	R-squared
Model 1	4.292	3.559	0.724
Model 2	4.323	3.593	0.745

Fig. 8. Cross validation results to compare candidate model performance

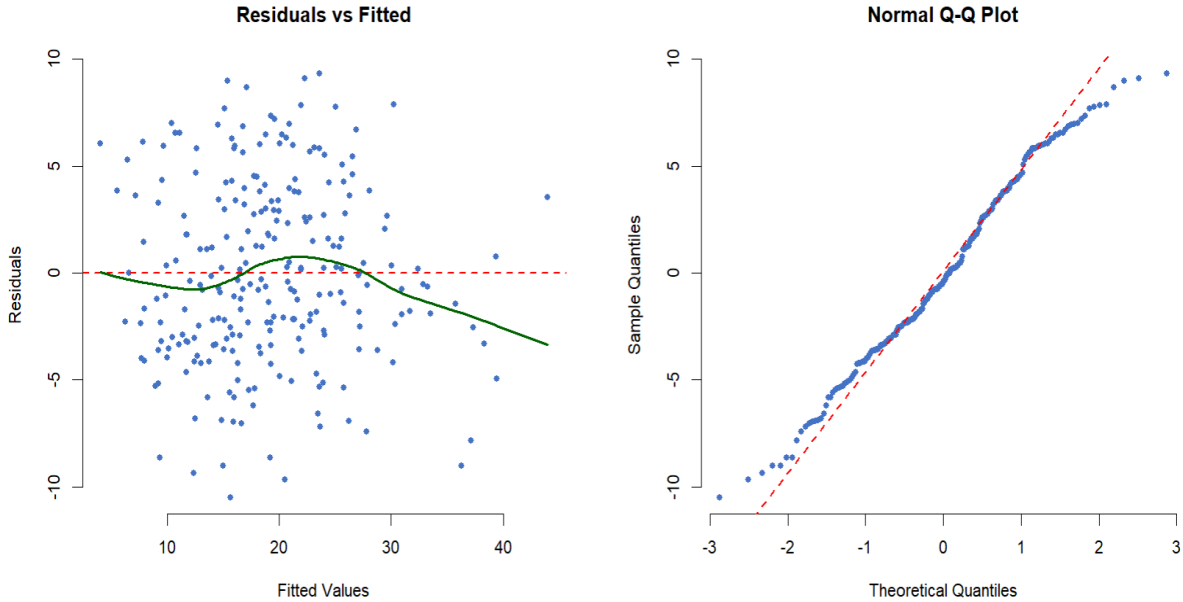

Fig. 9. Side-by-side diagnostic plots: residuals vs fitted values and normal Q-Q plot for final model

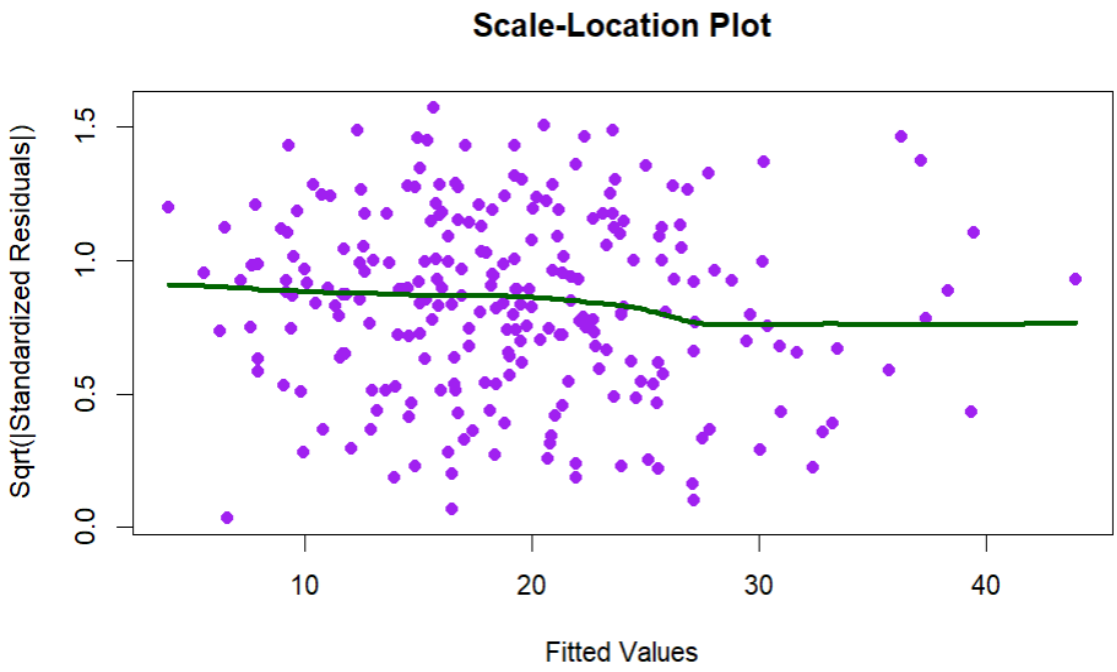

Fig. 10. Scale Location Plot to assess heteroskedasticity

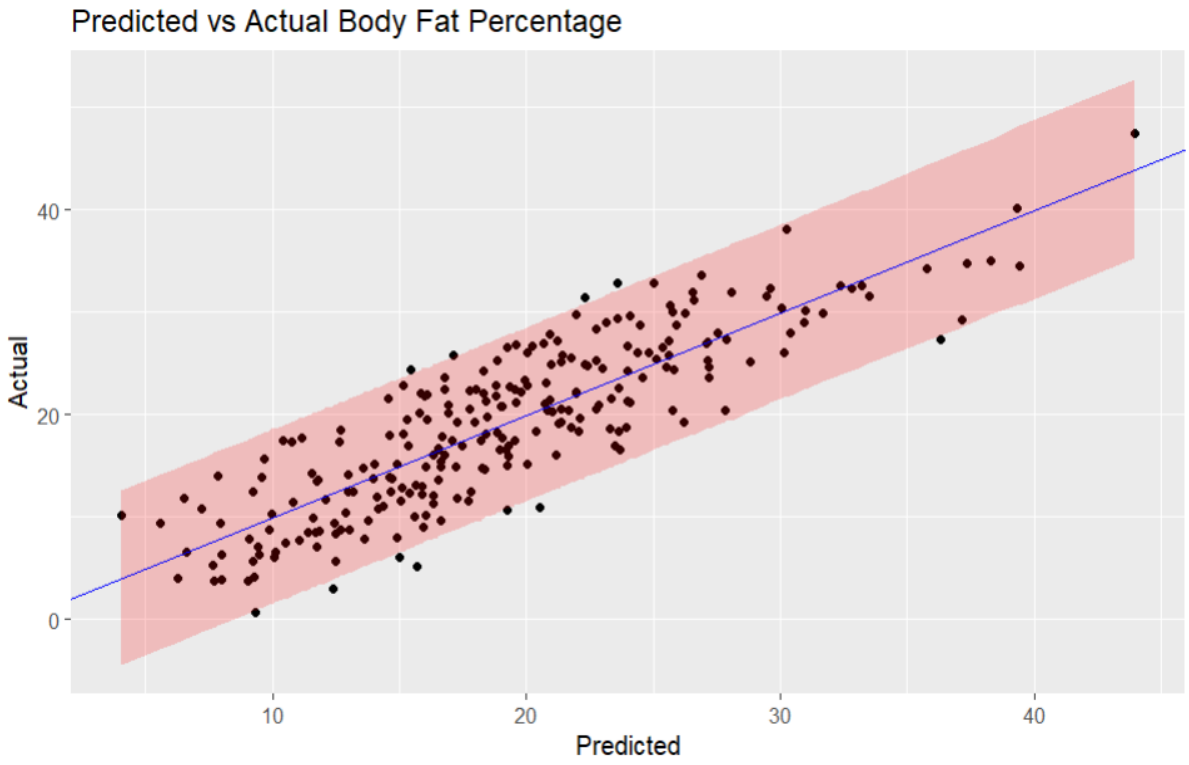

Fig. 11. 95% Prediction Interval